

SHEVANG PATEL

Long Beach, CA (open to relocation) | 562-234-4117 | shevangpatel0605@gmail.com | linkedin.com/in/shevangpatel | github.com/shevangpatel

EDUCATION

California State University – Long Beach

Master of Science in Computer Science | GPA: 3.7

Aug. 2023 – May. 2025

California, USA

Gujarat Technological University

Bachelor of Engineering in Information Technology | GPA: 3.48

Aug. 2018 – Jul. 2022

Ahmedabad, India

EXPERIENCE

Research Assistant | California State University — Long Beach, CA

Jan. 2025 – Present

- Built a **gait-analysis system** that reads smart-insole data (pressure sensors, accelerometer, gyroscope), segments each walk into fixed-length cycles, and prepares uniform tensors for stride-length prediction.
- Designed a custom **uncertainty loss** by combining Gaussian negative log-likelihood with Kullback–Leibler divergence, then ran a thorough hyperparameter search to discover the best balance which has lowest error between predicted and actual stride lengths.
- Evaluated the model with **10-fold cross-validation** on **37 subjects**, cutting average error by **12%** and variability by **30%**, and created intuitive Python dashboards to explore error patterns and confidence estimates.

Software Engineer | Skillcode Hub — Vadodara, India

Jul. 2022 – Jul. 2023

- Led development across the **Java and Spring boot backend** and **React and TypeScript frontend**, improving the fitness web platform's modules for onboarding, plan selection, trainer assignment, and posture feedback.
- Integrated the system with **AWS services** such as **EC2, S3, and RDS**, optimizing deployments, asset storage, and data management to achieve high reliability and scalability during user traffic spikes.
- Implemented and optimized **membership renewal, payment, and notification flows** with webhook retries and asynchronous updates, reducing payment errors by 30% and improving user retention.
- Strengthened **system security** and stability by implementing **token-based authentication**, API rate-limiting, and fine-grained access controls, minimizing unauthorized requests and improving reliability under stress.

Software Engineer Intern | Skillcode Hub — Vadodara, India

Jan. 2022 – Jun. 2022

- Built foundational modules for **user registration, login, trainer matching, and session booking** using **Spring Boot** for backend APIs and **React and TypeScript** for responsive frontend interfaces.
- Deployed prototype builds on **AWS (S3 and CloudFront)** and learned the fundamentals of CI/CD, environment management, and containerized application delivery. Contributed to initial versions of RBAC (Role-Based Access Control), **logging, and validation frameworks**, establishing coding standards and reliability practices for later production scaling.
- Worked closely with frontend and QA teams to define **REST API contracts**, align request/response structures, and implement integration testing, ensuring smooth cross-team collaboration.

PROJECTS

Resource Hub | Java, Spring Boot, React, TypeScript, PostgreSQL, AWS, Docker

- Built a **full-stack matching app** for assigning people to projects using a **Spring Boot API**, a **React and TypeScript web UI**, and **PostgreSQL** to store skills, availability, and project needs clearly.
- Implemented the matching logic with the **Hungarian algorithm**; exposed clean **REST endpoints** and a small **gRPC** service so other tools could request best matches quickly and reliably.
- Shipped with **Docker** and simple **CI/CD**; added basic metrics, error logs, and input validation cutting manual staffing time and making assignments more consistent across teams.

High-Efficiency Path Planning with Ray Tracing | Python, Dynamic Programming, Unity

- Developed an intelligent **pathfinding system** using **Dynamic Programming** to reach from starting to end point, capable of solving complex larger grid traversal problems with high memory efficiency and performance.
- Designed **cost functions**, enabling accurate and real-time obstacle detection across dynamic environments, achieving a **40% reduction** in computational time for complex pathfinding tasks and enhancing overall algorithm responsiveness.

AGENTIC RAG System for Document Retrieval | LangChain, Hugging Face, AstraDB, Python

- Built a GenAI-powered **RAG system** that answers user questions by first finding relevant text in both structured and unstructured documents using **semantic embeddings** and transformer-based search.
- Leveraged **AstraDB's** Cassandra vector storage and **custom tokenizer batching** to handle large-scale embedding lookups, delivering quick, low-latency responses even under heavy query loads.

TECHNICAL SKILLS

Languages: C# · Java · Python · Go · TypeScript · JavaScript · C++ · SQL || **Database:** PostgreSQL · MySQL · MongoDB

Backend: .NET · Spring Boot · Django · Gin · Node.js · GraphQL · JSON · JUnit · Mockito || **Frontend:** Angular · React · Webpack · Babel

Machine Learning: TensorFlow · NLP · LLM · PyTorch · Keras · OpenCV · Pandas · Numpy || **Project Management:** Jira · Agile · Scrum

Tools: AWS · Docker · Kubernetes · Terraform · Apache Kafka · Elasticsearch · Jenkins · Git · Linux · Gradle · Maven · Swagger · OpenAPI